- Early in a run, Predictors assign relatively optimistic probabilities to weaker Baselines (e.g., Gemma 2B, Haiku) on structurally complex questions (primes, high exponents, large LCMs).
- After observing several failures, they downgrade expectations and become quicker to bet that those models will be wrong, especially on questions that “look similar” in structure.

Table 5. Learning improvement (Round 4 – Round 1) by Predictor.

Predictor	Control R4−R1 (pp)	Incentive R4−R1 (pp)	Difference (pp)
Grok	0.0	10.6	+10.6
Mistral	4.3	11.1	+6.8
Qwen	4.4	14.2	+9.8
All	**2.9**	**12.0**	**+9.1**

Note. Each predictor value is the mean of 3 runs. pp = percentage points.

From Round 1 to Round 4, the Grok, Mistral, and Qwen Control models improved by 0.0%, 4.3%, and 4.4%, respectively. At the same time, those same models in the Incentive case improved by 10.6%, 11.1%, and 14.2%. Just one of the nine Incentive cases saw a decrease in performance between rounds, while four of the nine control cases were worse. In the Control group, a "Wrong" prediction is just a token. In the Incentive group, a "Wrong" prediction is accompanied by a numerical drop in bankroll. It could be that this numerical signal is much "louder" to the attention mechanism. The incentive structure created a steeper learning curve, allowing the models to adjust their weights (figuratively) much faster to the reality of the dataset.

This pattern supports H2: both conditions learn from feedback, and Incentive runs tend to produce steeper improvements; although H2 was statistically significant, we interpret it as an effect-size estimate in a small, exploratory pilot.

3.4 Bankroll dynamics and stake behavior

It was fascinating to see how the Incentive models **leveraged their stakes**.

Across all Incentive runs:

- No Predictor lost LLMCoin in any round.
- Final bankrolls were substantially above the starting 1,000,000 LLMCoin.

- Average ROI was **569%** (final bankroll ≈ **6.7M LLMCoin**), with a range from roughly 378% to 835% across individual runs.

Table 6. Bankroll dynamics across rounds (Incentive condition).

Predictor	After R1	After R2	After R3	Final (R4)
Grok-1	1.5M	2.3M	3.7M	5.8M
Grok-2	1.5M	2.4M	4.1M	6.1M
Grok-3	1.6M	2.8M	4.4M	7.1M
Mistral-1	1.6M	2.6M	4.2M	6.7M
Mistral-2	1.5M	2.2M	3.3M	5.0M
Mistral-3	1.6M	2.3M	3.4M	4.8M
Qwen-1	2.0M	3.9M	6.7M	9.1M
Qwen-2	1.0M	1.9M	3.4M	6.3M
Qwen-3	1.6M	2.8M	5.0M	9.3M
Average	**1.5M**	**2.6M**	**4.3M**	**6.7M**

Note. All runs began with 1.0M LLMCoin. M = millions. The average final bankroll of 6.7M represents a 569% ROI (i.e., profit relative to the initial 1.0M bankroll).

Stake size and accuracy tracked each other closely. Grouping bets by stake magnitude:

- **Tiny stakes** of 1-1,000 LLMCoin corresponded to **mid-range accuracy (74%)**.
- **Mid-sized stakes** of 1,001-10,000 were slightly more accurate, around **76%**.
- **Large stakes** (10,000-40,000 coins) were **rarely wrong (~89%)**. And for **Whale Stakes**, bets 40,000 coins and higher (n = 170), only 2 of those were incorrect (**99% accuracy**).

Table 7. Stake size and prediction accuracy (Incentive condition).

Stake range	n bets	Correct	Accuracy (%)
1-1,000 (tiny)	871	644	73.9
1,001-10,000 (mid)	2,249	1,710	76.0
10,001-40,000 (large)	2,110	1,879	89.1
≥40,000 (whale)	**170**	**168**	**98.8**

Note. Stakes ranged from 1 to 100,000 LLMCoin per prediction. Whale bets (≥40,000) showed near-perfect accuracy.

In other words, when the model "really cared" about being right, it was almost always right. The betting mechanic created a **clean, monotonic confidence channel** that is largely absent in standard yes/no outputs.

Predictor-specific behavior:

- **Qwen 3 Max Thinking** behaved most like a disciplined Kelly-style bettor (sizing bets proportionally to edge), concentrating large bets on structurally easy items for strong Baselines and using tiny hedges where it was unsure. Yet in the first round of its second run, Qwen showed the most conservative betting of any model, staking the **minimum 1 LLMCoin on every question** before ramping up in Round 2.
- **Grok 4 Expert** and **Mistral Large 2** also used the full stake range but displayed more variance, with occasional over- or undersized bets that did not align perfectly with actual difficulty.

3.5 Arithmetic fragility in the incentive runs

In three Incentive runs (two Mistral, one Grok), the model slightly exceeded the per-round bankroll constraint—on the order of **~0.2% overspend**—while still tracking wins and losses correctly. These violations did not change the binary Yes/No predictions, only the wager magnitude.

This suggests a subtle failure mode:

- The model's **internal ledger** of wins and losses was coherent.
- But its **planning arithmetic** for future stakes (summing proposed bets and comparing to bankroll) was slightly off.

From a forecasting perspective, the underlying *predictions* remain valid; the ROI is mildly inflated relative to a perfectly constraint-respecting strategy. More broadly, it highlights that:

LLMs can strategize financially, but basic arithmetic checks should be enforced *outside* the model rather than relying on the model to police its own constraints.

3.6 RQ3 / H3: Predicting current vs prior-generation models

The preregistration called for comparing forecasting accuracy across Baseline tiers: Table 8 summarizes tier-level prediction accuracy by condition.